

LaTeX Test Cases

Test Suite

January 25, 2026

1 Basic Test Cases

This is a simple paragraph with some text to test basic formatting.

Here's an equation: $E = mc^2$

1.1 Lists

- First item
- Second item
- Third item
- Fourth item

2 Complicated Test Cases

2.1 Complex Mathematical Expression

Consider the following integral:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx = 1$$

And a more complex matrix operation:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

2.2 Nested Structures

1. Outer item one
 - Inner item a
 - Inner item b
 - (a) Deeply nested item i
 - (b) Deeply nested item ii
2. Outer item two

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x+1)^2 \end{aligned}$$

2.3 Advanced Formatting

Bold text and *italic text* and monospaced `text`.

Here's a block with special characters: \$, #, %, &, -, {, }

3 Conclusion

This document tests various LaTeX features and should render correctly.